

OPLOGS-05: Querying & Parsing Logs

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DQL Fundamentals and DPL Pattern Matching

This notebook covers DQL query syntax, filtering, string matching, and DPL (Dynatrace Pattern Language) for extracting structured data from logs.

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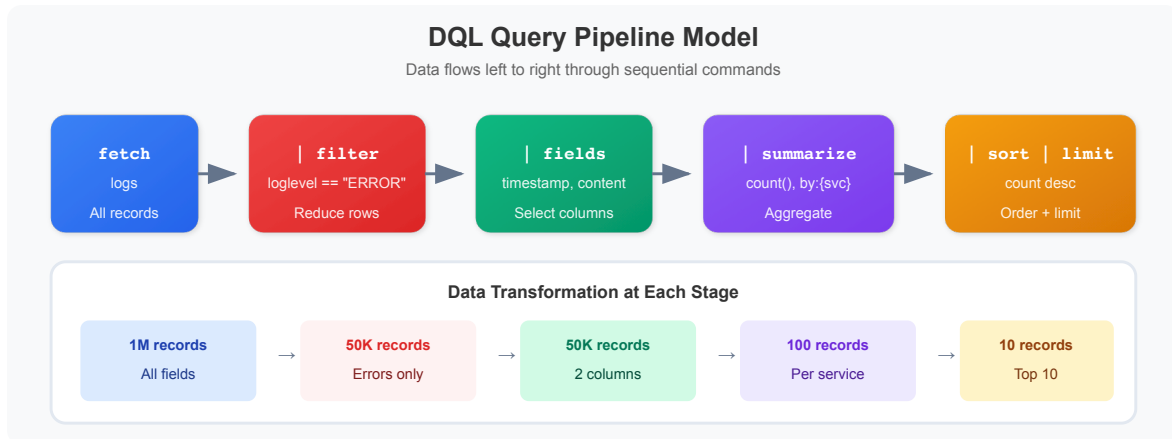
Prerequisites

-  Access to a Dynatrace environment with log data

-  Completed OPLOGS-01 through OPLOGS-04
-  Basic understanding of regular expressions (helpful)

1. DQL Fundamentals

Query Structure



```

fetch logs, from: now() - 1h           // Data source and time range
| filter <condition>                    // Filter records
| fields <field1>, <field2>              // Select fields
| sort <field> desc                     // Order results
| limit 100                            // Limit output
  
```

Critical DQL Rules (DQL ≠ SQL)

 DQL Syntax	 SQL Syntax
<code>filter status == "ERROR"</code>	<code>WHERE status = 'ERROR'</code>
<code>in(field, {"a", "b"})</code>	<code>field IN ('a', 'b')</code>
<code>summarize {count = count()}</code>	<code>SELECT COUNT(*)</code>
<code>isNull(field)</code>	<code>field IS NULL</code>

```

// Basic log query - explore recent logs
fetch logs, from: now() - 1h
| fields timestamp, content, loglevel, dt.entity.host
| limit 10
  
```

```
// Query with time range options
fetch logs, from: now() - 24h, to: now() - 12h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

2. Filtering Logs

Comparison Operators

Operator	Description	Example
<code>==</code>	Equals	<code>loglevel == "ERROR"</code>
<code>!=</code>	Not equals	<code>status != "NONE"</code>
<code>></code> , <code>>=</code>	Greater than	<code>count > 100</code>
<code><</code> , <code><=</code>	Less than	<code>duration < 1000</code>

Logical Operators

Operator	Description	Example
<code>AND</code>	Both conditions	<code>loglevel == "ERROR" AND isNotNull(host)</code>
<code>OR</code>	Either condition	<code>loglevel == "ERROR" OR loglevel == "WARN"</code>
<code>NOT</code>	Negate	<code>NOT contains(content, "health")</code>

```
// Filter by log level
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fields timestamp, content, dt.entity.host
| limit 20
```

```
// Multiple conditions with AND/OR
fetch logs, from: now() - 1h
| filter (loglevel == "ERROR" OR loglevel == "WARN")
      AND isNotNull(dt.entity.host)
| summarize {count = count()}, by: {loglevel, dt.entity.host}
| sort count desc
| limit 15
```

```
// Filter using in() for multiple values
fetch logs, from: now() - 1h
| filter in(loglevel, {"ERROR", "WARN", "INFO"})
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

3. String Matching Functions

Function	Description	Example
<code>contains(str, substr)</code>	Substring match	<code>contains(content, "error")</code>
<code>startsWith(str, prefix)</code>	Prefix match	<code>startsWith(content, "[ERROR]")</code>
<code>endsWith(str, suffix)</code>	Suffix match	<code>endsWith(content, "failed")</code>
<code>matchesPhrase(str, phrase)</code>	Word boundary match	<code>matchesPhrase(content, "connection refused")</code>
<code>matchesValue(str, pattern)</code>	Exact or wildcard match	<code>matchesValue(host, "web-*")</code>

```
// Search for specific content patterns
fetch logs, from: now() - 1h
| filter contains(content, "Exception")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| fields timestamp, content_preview, dt.entity.host
| limit 15
```

```
// Use matchesPhrase for word-boundary matching
fetch logs, from: now() - 1h
| filter matchesPhrase(content, "connection")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

```
// Exclude patterns with NOT
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| filter NOT matchesPhrase(content, "health")
| filter NOT matchesPhrase(content, "heartbeat")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

4. Field Selection and Transformation

Field Commands

Command	Description
<code>fields</code>	Select specific fields only
<code>fieldsAdd</code>	Add computed fields
<code>fieldsRemove</code>	Remove fields

```
// Add computed fields
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fieldsAdd severity = if(contains(content, "critical"), "CRITICAL",
                        else: if(contains(content, "fatal"), "FATAL",
                                else: "ERROR"))
| fieldsAdd content_length = stringLength(content)
| fields timestamp, severity, content_length, dt.entity.host
| limit 20
```

```
// String manipulation functions
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd pod_short = substring(k8s.pod.name, from: 0, to: 30)
| fields timestamp, k8s.namespace.name, pod_short, loglevel
| limit 10
```

5. DPL (Dynatrace Pattern Language) Parsing

DPL extracts structured data from unstructured log content.

DPL Matchers

Matcher	Description	Matches
INT	Integer	42 , -17
DOUBLE	Decimal	3.14 , -0.5
IPADDR	IP address	192.168.1.1 , ::1
LD	Line data (to delimiter)	Any text
WORD	Word characters	hello , user123
SPACE	Whitespace	spaces, tabs
JSON	JSON structure	{"key": "value"}

Pattern Syntax

Syntax	Description
MATCHER:fieldname	Extract to named field
MATCHER	Match but don't extract
MATCHER?	Optional matcher
'literal'	Match exact string
(opt1 opt2)	Match alternatives

```
// Parse log level from content (e.g., "[ERROR] message")
fetch logs, from: now() - 1h
| filter startsWith(content, "[")
| parse content, "'[' LD:parsed_level ']"
| filter isNotNull(parsed_level)
| summarize {count = count()}, by: {parsed_level}
| sort count desc
| limit 10
```

```
// Parse HTTP-style logs
// Example: GET /api/users 200 45ms
fetch logs, from: now() - 1h
| filter contains(content, "GET") OR contains(content, "POST")
| parse content, "LD:method SPACE '/' LD:path SPACE INT:status_code"
| filter isNotNull(status_code)
| summarize {count = count()}, by: {method, status_code}
| sort count desc
| limit 15
```

```
// Parse key-value pairs
// Example: "user=john action=login status=success"
fetch logs, from: now() - 1h
| parse content, "'user=' LD:username SPACE"
| filter isNotNull(username)
| summarize {count = count()}, by: {username}
| sort count desc
| limit 10
```

```
// Parse IP addresses from logs
fetch logs, from: now() - 1h
| parse content, "IPADDR:client_ip"
| filter isNotNull(client_ip)
| summarize {count = count()}, by: {client_ip}
| sort count desc
| limit 15
```

6. Working with JSON Logs

Many applications emit JSON-formatted logs. DQL provides tools to work with JSON content.

```
// Find JSON-formatted logs
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| fieldsAdd content_preview = substring(content, from: 0, to: 150)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 10
```

```
// Parse JSON and extract fields
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| parse content, "JSON:json_data"
| filter isNotNull(json_data)
| fields timestamp, json_data
| limit 10
```

```
// Access nested JSON fields
fetch logs, from: now() - 1h
| filter startsWith(content, "{")
| parse content, "JSON:json_data"
| filter isNotNull(json_data)
| fieldsAdd error_type = json_data[errorType]
| fieldsAdd status_val = json_data[status]
| filter isNotNull(status_val)
| summarize {count = count()}, by: {status_val, error_type}
| sort count desc
| limit 15
```

7. Null Handling

DQL uses three-valued logic. `NULL` comparisons require special functions.

❌ Wrong	✅ Correct
<code>field == null</code>	<code>isNull(field)</code>
<code>field != null</code>	<code>isNotNull(field)</code>


```
// Check for null values
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    with_host = countIf(isNotNull(dt.entity.host)),
    without_host = countIf(isNull(dt.entity.host)),
    with_namespace = countIf(isNotNull(k8s.namespace.name)),
    without_namespace = countIf(isNull(k8s.namespace.name))
}
```

```
// Use coalesce for default values
fetch logs, from: now() - 1h
| fieldsAdd effective_level = coalesce(loglevel, status, "UNKNOWN")
| summarize {count = count()}, by: {effective_level}
| sort count desc
```

8. Advanced Parsing Examples

```
// Parse exception patterns
fetch logs, from: now() - 24h
| filter contains(content, "Exception") OR contains(content, "Error")
| parse content, "LD:exception_type 'Exception'"
| filter isNotNull(exception_type)
| summarize {count = count()}, by: {exception_type}
| sort count desc
| limit 15
```

```
// Parse with optional fields
// Matches: "error code=123" or "error code=123 message=failed"
fetch logs, from: now() - 1h
| filter contains(content, "error")
| parse content, "'error' (SPACE 'code=' INT:error_code)? (SPACE 'message='
LD:error_msg)?"
| filter isNotNull(error_code)
| summarize {count = count()}, by: {error_code}
| sort count desc
| limit 10
```

```
// Parse alternative formats
// Matches: "user=john", "username=john"
fetch logs, from: now() - 1h
| parse content, "('user='|'username=') LD:user_value"
| filter isNotNull(user_value)
| summarize {count = count()}, by: {user_value}
| sort count desc
| limit 10
```



Summary

In this notebook, you learned:

- ✓ **DQL fundamentals** - Query structure and syntax rules
- ✓ **Filtering** - Comparison operators, logical operators, in()
- ✓ **String matching** - contains, matchesPhrase, startsWith
- ✓ **Field manipulation** - fieldsAdd, computed fields
- ✓ **DPL parsing** - Matchers (INT, LD, IPADDR, JSON)
- ✓ **JSON handling** - Parsing and accessing nested fields
- ✓ **Null handling** - isNull, isNotNull, coalesce



Next Steps

Continue to **OPLOGS-06: Topology & Entity Context** to learn about entity context and relationships.



References

- [DQL Reference](#)
- [DQL Functions](#)
- [Dynatrace Pattern Language](#)
- [DPL Architect Tool](#)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.